

Dirichlet Multinomial Inference

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1 Dirichlet Multinomial Inference

Now we shall generalize the Beta-Binomial analysis to Dirichlet-Multinomial inference. First we shall look at the Multinomial distribution (which is a generalization of the Binomial distribution) and the Dirichlet distribution (which is a generalization of the Beta distribution).

1.1 Multinomial Distribution

The multinomial distribution is a generalization of the Binomial distribution. We first recall the binomial distribution. We say that $X \sim \text{Bin}(n, p)$ if

$$P\{X = x\} = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \quad \text{for } x = 0, 1, \dots, n.$$

X represents the count of success in n repetitions of a simple binary experiment with two outcomes success (probability p) and failure (probability $1-p$).

The multinomial distribution is obtained by considering n repetitions of an experiment with k outcomes (for some $k \geq 1$) with probabilities $p_1, \dots, p_k$ (we need $p_i \geq 0$ and $\sum_{i=1}^k p_i = 1$). Let X_i denote the count of the i -th outcome among the n repetitions of the experiment (in other words, X_i is the number of times the i -th outcome happens in the n trials). The joint distribution of $(X_1, \dots, X_k)$ is written as Multinomial($n; p_1, \dots, p_k$). It corresponds to the probabilities:

$$P\{X_1 = x_1, \dots, X_k = x_k\} = \begin{cases} \frac{n!}{x_1! \dots x_k!} \prod_{i=1}^k p_i^{x_i}, & \text{if } x_i \geq 0 \text{ for all } i \text{ and } \sum_{i=1}^k x_i = n, \\ 0, & \text{otherwise.} \end{cases}$$

The following consequences of $(X_1, \dots, X_k) \sim \text{Multinomial}(n; p_1, \dots, p_k)$ are straightforward to see:

1. To derive the distribution of X_i i.e., $P\{X_i = x\}$, we can write

$$\begin{aligned} P\{X_i = x\} &= P\{x \text{ trials had outcome } i, (n-x) \text{ trials had outcome not } i\} \\ &= \frac{n!}{x!(n-x)!} p_i^x (1-p_i)^{n-x}. \end{aligned}$$

In other words $X_i \sim \text{Bin}(n, p_i)$. In essence, here we consider the variant of the original experiment where, for each trial, we just record if the original outcome was 'not i ' or i . This converts the setup to binary trials and X_i represents the number of successes (success = outcome is i) so $X_i \sim \text{Bin}(n, p_i)$.

2. The joint distribution of (X_i, X_j) (for fixed $i \neq j$) can be derived in the following way:

$$\begin{aligned} P\{X_i = x_1, X_j = x_2\} &= P\{x_1 \text{ trials were } i, x_2 \text{ trials were } j, n - x_1 - x_2 \text{ trials were neither } i \text{ nor } j\} \\ &= \frac{n!}{x_1! x_2! (n - x_1 - x_2)!} p_i^{x_1} p_j^{x_2} (1 - p_i - p_j)^{n - x_1 - x_2} \end{aligned}$$

provided $x_1, x_2 \in \{0, 1, \dots, n\}$ with $x_1 + x_2 \leq n$ (otherwise, the probability equals 0).

3. It can be checked (see e.g., https://en.wikipedia.org/wiki/Multinomial_distribution) that

$$\mathbb{E}X_i = np_i, \quad \text{var}(X_i) = np_i(1-p_i), \quad \text{and} \quad \text{cov}(X_i, X_j) = -np_i p_j \text{ if } i \neq j.$$

1.2 Dirichlet Distribution

The Dirichlet distribution is a generalization of the Beta distribution, see Gelman et al. [2013, Chapter 3]). Recall the Beta(a, b) distribution is a distribution over $p \in [0, 1]$ corresponding to the density:

$$f(p) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} p^{a-1} (1-p)^{b-1} I\{0 \leq p \leq 1\}.$$

This density function is well-defined only when both a and b are strictly positive. However one can still define the Beta(a, b) distribution even when one or both of a and b are zero. This can be done by a limiting argument:

1. Beta($0, 0$) = $\lim_{\epsilon \downarrow 0} \text{Beta}(\epsilon, \epsilon) = \text{Bernoulli}(0.5)$. In other words, Beta($0, 0$) becomes a discrete two-point distribution assigning equal probability to 0 and 1.
2. For fixed $b > 0$, we have Beta($0, b$) = $\lim_{\epsilon \downarrow 0} \text{Beta}(\epsilon, b) = \text{Bernoulli}(0) = \delta_{\{0\}}$. In other words, Beta($0, b$) is the point mass as 0.
3. For fixed $a > 0$, we have Beta($a, 0$) = $\lim_{\epsilon \downarrow 0} \text{Beta}(a, \epsilon) = \text{Bernoulli}(1) = \delta_{\{1\}}$. In other words, Beta($a, 0$) is the point mass as 1.

The limiting statements above can be made rigorous by moment calculations (e.g., by showing that the moments of Beta(ϵ, ϵ) converge to the moments of Bernoulli(0.5) as $\epsilon \downarrow 0$).

The Beta distribution can be viewed as the distribution over the probabilities (p and $1-p$) of an experiment with two outcomes. The Dirichlet distribution is the distribution over the probabilities $p_1, \dots, p_k$ of an experiment with k outcomes. Specifically given $a_1, \dots, a_k$, the Dirichlet($a_1, \dots, a_k$) distribution corresponds to the density:

$$\frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} p_1^{a_1-1} \dots p_k^{a_k-1} I\{p_1, \dots, p_k \geq 0, \sum_i p_i = 1\}.$$

Note that this should be viewed as density on $(k-1)$ -dimensional space (as opposed to k -dimensional space) because of the restriction $p_1 + \dots + p_k = 1$. In other words, for every $A \subseteq^k$, we have

$$\begin{aligned} & P\{(p_1, \dots, p_k) \in A\} \\ &= \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \int_S p_1^{a_1-1} \dots p_{k-1}^{a_{k-1}-1} (1 - p_1 - \dots - p_{k-1})^{a_k-1} \\ & I\{(p_1, \dots, p_{k-1}, 1 - p_1 - \dots - p_{k-1}) \in A\} dp_1 \dots dp_{k-1} \end{aligned}$$

where

$$S := \{(p_1, \dots, p_{k-1}) : p_i \geq 0, p_1 + \dots + p_{k-1} \leq 1\}.$$

It can be checked that if $(p_1, \dots, p_k) \sim \text{Dirichlet}(a_1, \dots, a_k)$, then (see e.g., https://en.wikipedia.org/wiki/Dirichlet_distribution)

$$\begin{aligned} \mathbb{E}p_i &= \frac{a_i}{a_1 + \dots + a_k} \\ \text{var}(p_i) &= \left(\frac{a_i}{a_1 + \dots + a_k} \right) \left(\frac{(a_1 + \dots + a_k) - a_i}{a_1 + \dots + a_k} \right) \left(\frac{1}{a_1 + \dots + a_k + 1} \right) \\ \text{cov}(p_i, p_j) &= - \left(\frac{a_i}{a_1 + \dots + a_k} \right) \left(\frac{a_j}{a_1 + \dots + a_k} \right) \left(\frac{1}{a_1 + \dots + a_k + 1} \right). \end{aligned}$$

*What does this mean??
Can I enclose the simplex within a box; then define the density 0 outside the box??*

Remark 1 (Connection with the Gamma distribution). *A particularly useful way to construct the Dirichlet distribution is through independent Gamma random variables. Let*

$$G_i \stackrel{\text{ind}}{\sim} \text{Gamma}(a_i, 1), \quad i = 1, \dots, k,$$

where the Gamma density is

$$f(x) = \frac{1}{\Gamma(a_i)} x^{a_i-1} e^{-x}, \quad x > 0.$$

Define

$$T = G_1 + \dots + G_k, \quad P_i = \frac{G_i}{T}, \quad i = 1, \dots, k.$$

Then

$$(P_1, \dots, P_k) \sim \text{Dirichlet}(a_1, \dots, a_k),$$

and the random variable T is independent of $(P_1, \dots, P_k)$ with

$$T \sim \text{Gamma}(a_1 + \dots + a_k, 1).$$

Thus the Dirichlet distribution can be interpreted as the distribution obtained by taking independent Gamma random variables and normalizing them so that they sum to one. This representation explains why the Dirichlet distribution is supported on the probability simplex, and is also the basis of the simplest algorithm for generating Dirichlet random vectors in practice. The Beta distribution is the special case $k = 2$. Indeed, if

$$G_1 \sim \text{Gamma}(a, 1), \quad G_2 \sim \text{Gamma}(b, 1),$$

are independent, then

$$\frac{G_1}{G_1 + G_2} \sim \text{Beta}(a, b).$$

As for the case of the Beta, the following facts about the Dirichlet can also be proved using moment calculations:

1. $\text{Dirichlet}(0, \dots, 0) = \lim_{\epsilon \downarrow 0} \text{Dirichlet}(\epsilon, \dots, \epsilon)$ is the discrete uniform distribution on $e_1, \dots, e_k$ with e_i is the vector which has 1 in the i -th location and 0 everywhere else i.e.,

$$\text{Dirichlet}(0, \dots, 0) = \frac{1}{k} \sum_{i=1}^k \delta_{\{e_i\}}. \quad \rightarrow \text{not obvious!}$$

2. If $k = 4$ and $a_2, a_4 > 0$, then $\text{Dirichlet}(0, a_2, 0, a_4) = \lim_{\epsilon \downarrow 0} \text{Dirichlet}(\epsilon, a_2, \epsilon, a_4)$ is supported on $\{(p_1, p_2, p_3, p_4) : p_1 = 0, p_2 \geq 0, p_3 = 0, p_4 \geq 0, p_1 + p_2 + p_3 + p_4 = 1\}$, and the distribution of (p_2, p_4) is $\text{Dirichlet}(a_2, a_4)$. Informally, we write

$$\text{Dirichlet}(0, a_2, 0, a_4) = \text{Dirichlet}(a_2, a_4).$$

3. More generally, let $a = (a_1, \dots, a_k)$ with each $a_i \geq 0$. Let $I = \{i : a_i > 0\}$ and let m be the cardinality of I . Then $\text{Dirichlet}(a) := \lim_{\epsilon \downarrow 0} \text{Dirichlet}(a_1 + \epsilon, \dots, a_k + \epsilon)$ satisfies the following.

- (a) The support of $\text{Dirichlet}(a)$ equals $\{(p_1, \dots, p_k) : p_i \geq 0, \sum_{i=1}^k p_i = 1, p_i = 0 \text{ for all } i \notin I\}$.
- (b) If $p \sim \text{Dirichlet}(a)$, then the subvector $p_i, i \in I$ satisfies:

$$(p_i, i \in I) \sim \text{Dirichlet}(a_i, i \in I).$$

Note that the above is a Dirichlet distribution for an m -dimensional probability vector.

1.3 Dirichlet Prior and Multinomial Likelihood

We saw the following basic fact:

$$p \sim \text{Beta}(a, b) \quad \text{and} \quad X | p \sim \text{Bin}(n, p) \implies p | X = x \sim \text{Beta}(a + x, b + n - x).$$

The generalization of this is:

$$(p_1, \dots, p_k) \sim \text{Dirichlet}(a_1, \dots, a_k) \quad \text{and} \quad (X_1, \dots, X_k) | (p_1, \dots, p_k) \sim \text{Multinomial}(n; p_1, \dots, p_k) \\ \implies (p_1, \dots, p_k) | X_1 = x_1, \dots, X_k = x_k \sim \text{Dirichlet}(a_1 + x_1, a_2 + x_2, \dots, a_k + x_k).$$

So the posterior mean estimate of p_i is given by:

$$\mathbb{E}(p_i | X_1 = x_1, \dots, X_k = x_k) = \frac{a_i + x_i}{(a_1 + x_1) + \dots + (a_k + x_k)} = \frac{a_i + x_i}{n + a_1 + \dots + a_k}$$

where we used $x_1 + \dots + x_k = n$.

Remark 2. The above posterior characterization is true for any $a_1, \dots, a_k$ and then one can also make sense of the statement for all $a_1, \dots, a_k$ equal to 0 by taking limits.

1.4 Special Case: the Dirichlet(0, ..., 0) prior

In the special case where all $a_i = 0$, then the posterior mean of p_i is simply x_i/n ; so we are estimating the probability of the i -th outcome by simply the proportion of times the i -th outcome appeared in the n trials. This is just the frequentist MLE. So the posterior mean estimate coincides with the frequentist MLE when the prior is Dirichlet(0, ..., 0).

More precisely, the posterior for the Dirichlet(0, ..., 0) prior is Dirichlet($x_1, \dots, x_k$). A key property of this posterior is that if $x_i = 0$ for some i , then the posterior places all its mass on the set $\{p_i = 0\}$. In other words, any outcome that is not observed in the sample is assigned zero probability under the posterior and is therefore completely excluded from future consideration.

As an explicit example, suppose $k = 6$, $n = 4$, $x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 0, x_5 = 3, x_6 = 1$. In other words, we rolled a six-faced die 4 times; in 3 of the 4 rolls, we observed the outcome 5, and in the other roll, we observed a 6. After observing this data (note that we started with the prior Dirichlet(0, 0, 0, 0, 0, 0)), our posterior becomes Dirichlet(0, 0, 0, 0, 3, 1). This means that the posterior is concentrated on the probabilities $(0, 0, 0, 0, p_5, 1 - p_5)$ with $p_5 \sim \text{Beta}(3, 1)$.

Remark 3 (Interpretation of the Dirichlet(0, ..., 0) prior). The prior Dirichlet(0, ..., 0) has the attractive property that the posterior mean coincides exactly with the frequentist maximum likelihood estimator:

$$E(p_i | X) = \frac{x_i}{n}.$$

Thus the prior contributes no pseudo-counts to the posterior and, in this limited sense, behaves as an “uninformative” prior.

It is important to note, however, that this prior is not noninformative in the usual Bayesian sense. Indeed,

$$\text{Dirichlet}(0, \dots, 0) = \lim_{\varepsilon \downarrow 0} \text{Dirichlet}(\varepsilon, \dots, \varepsilon)$$

is the discrete uniform distribution on the vertices of the probability simplex. Thus, before observing any data, the prior assigns equal probability to the hypotheses

$$(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, \dots, 0, 1),$$

Bayesian Bootstrap

Why Dirichlet (0) prior?

Jayno $\rightarrow$ Jar with white substance. We taste once, its sweet $\rightarrow$ sugar, not salt.

$p \rightarrow$ Proportion of sugar in the jar.

Prior $\rightarrow$ Either fully sugar or fully salt;

$$\text{Beta}(0,0) = \lim_{\epsilon \rightarrow 0} \text{Beta}(\epsilon, \epsilon).$$

Posterior $\text{Beta}(1,0) \rightarrow$ pt. mass at 1.

Taste it again, salt $\rightarrow$ now no idea.

Posterior $p \rightarrow \text{Beta}(1,1) \rightarrow$ Uniform

In general, posterior $\rightarrow$ Empirical Dist. on the observations $\rightarrow$ classical frequentist object!

Say we want to do inference on some

functional θ like the mean $\sum a_i p_i$ or
median $\left(\begin{array}{l} a_1 \rightarrow p_1 \\ \vdots \\ a_k \rightarrow p_k \end{array} \right)$.

Algorithm

Sample $(w_1, \dots, w_n) \sim \text{Dir}(1, \dots, 1)$

& calculate mean or median: M times

where M is large. This approximates the

posterior dist of θ . Derive $(1-\alpha)$

credible intervals from the histogram.

This algo is called Bayesian Bootstrap.

Bootstrap

1) X, Y r.v's with equal mean say.

Want to min.

$$\text{Var}(\alpha X + (1-\alpha)Y).$$

$$\alpha = \frac{\sigma_y^2 - \sigma_{xy}}{\sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}}.$$

Given (X_i, Y_i) i.i.d

can obtain $\hat{\alpha}$. How to estimate $\text{Var}(\hat{\alpha})$?

$$\hat{\alpha} = \frac{\hat{\sigma}_y^2 - \hat{\sigma}_{xy}}{\hat{\sigma}_x^2 + \hat{\sigma}_y^2 - 2\hat{\sigma}_{xy}}.$$

2) Generally I have some statistic

$X_1, \dots, X_n \text{ i.i.d } P$

$$T = T(X_1, \dots, X_n).$$

such as sample mean, median, reg-co-eff est.

How to estimate $\text{Var } T$??

3) Look at $\text{Var } T = f(P).$

Estimate P by $P_n \rightarrow$ If n is large, P_n is a good est. of P .

Arg-in principle

$$\hat{\text{Var } T} = f(P_n) = \text{Var } T(Y_1, \dots, Y_n)$$

where $Y_i \sim \text{i.i.d } P_n$.

4) Dists of $T(Y_1, \dots, Y_n)$ not available in closed form. $\downarrow$ discrete r.v. n^n possibilities.
Use Monte-Carlo.

5) Why bootstrap can work?

$f(P_n) \approx f(P)$. 1) P_n close to P (frequentist justification)

2) f is smooth functional.

Works well for mean, quantiles, corr. coeffs, regression coeffs..

6) Bootstrap can fail if f is not smooth.

One classical example is $T = \max(X_1, \dots, X_n)$.

Say $X_1, \dots, X_n$ i.i.d $U(0,1)$.

We want to estimate $\text{Var } T$.

Similarities of both Bootstraps

Sample $\rightarrow X_1, \dots, X_n$.

Want to estimate $E f(X_1, \dots, X_n) \leftrightarrow$ like
variance of sample median.

Usual Bootstrap

Sample $Y_1, \dots, Y_n$ n times w. replacement
or i.i.d P_n .

or equivalently, (random)
Create the distribution

$$\left[X_i \sim \frac{1}{n} \text{Mult} \left(n; \frac{1}{n}, \dots, \frac{1}{n} \right) \right]$$

Evaluate $E f$
 $\downarrow$
random distribution supported on the
sample.

Bayesian Bootstrap:

Same except $\frac{1}{n} \text{ Mult} \left(n; \frac{1}{n}, \dots, \frac{1}{n} \right)$

is replaced by $(w_1, \dots, w_n) \sim \text{Dir}(1, \dots, 1)$.

Evaluate

$E f$

↓
random distribution

supported on the
sample.

4 Simple Language Models

We will illustrate Dirichlet-Multinomial inference on a text data example. This analysis is taken from [MacKay and Peto \[1995\]](#). We observe some text $y_1, \dots, y_n$. The vocabulary consists of the number of distinct words (including punctuation marks) in the text. Assume the vocabulary size is k .

4.1 A simple i.i.d model

The simplest model assumes that $y_1, \dots, y_n$ are i.i.d with an distribution P that is supported on the vocabulary. The likelihood then becomes:

$$\prod_{i=1}^n p_{y_i} = \prod_{i=1}^k p_i^{N_i} \quad (14)$$

where N_i is the observed count for the i -th word in the vocabulary. We can place a Dirichlet prior on $(p_1, \dots, p_k)$ (e.g., $\text{Dirichlet}(0, \dots, 0)$) which would result in the $\text{Dirichlet}(N_1, \dots, N_k)$ posterior distribution. One can then generate new text in the following way:

1. First generate $(p_1, \dots, p_k)$ from the posterior distribution.
2. Then generate new words $y_{n+1}, y_{n+2}, \dots$ i.i.d from the multinomial distribution $\text{Mult}(1; p_1, \dots, p_k)$ until a specific word (e.g., the period symbol '.') is generated.

Obviously, this will not result in realistic text. This is because the i.i.d assumption collapses the text into a bag-of-words representation, discarding all sequential information. Indeed, in this model, we are effectively working with the counts $N_1, \dots, N_k$ instead of the raw data $y_1, \dots, y_n$. The likelihood in terms of the counts is:

$$\frac{n}{N_1! \dots N_k!} p_1^{N_1} \dots p_k^{N_k} \propto \prod_{i=1}^k p_i^{N_i}$$

which coincides with (14). As a result, it cannot capture even the most basic linguistic dependencies (e.g., that “the” is likely to be followed by a noun).

4.2 AR(1) model

A slightly better model will be obtained if we replace the i.i.d assumption with a Markovian one. This is also referred to as the first order AutoRegressive AR(1) model.

The likelihood here then becomes:

$$p_{y_1} p_{y_2|y_1} p_{y_3|y_2, y_1} \dots p_{y_n|y_{n-1}, \dots, y_1}$$

We assume now that $p_{y_i|y_{i-1}, \dots, y_1} = p_{y_i|y_{i-1}}$ for each $i = 3, \dots, n$. This gives

$$\text{Likelihood} = p_{y_1} \prod_{i=2}^n p_{y_i|y_{i-1}}.$$

We also assume that p_{y_1} is constant so we don't need to model it. This gives:

$$\text{Likelihood} \propto \prod_{i=2}^n p_{y_i|y_{i-1}}. \quad (15)$$

If you don't like the assumption that p_{y_1} is constant, you can interpret the above as the conditional likelihood of $(y_2, \dots, y_n)$ given y_1 .

Just as in the i.i.d model where we could write the likelihood in terms of counts, we can rewrite the likelihood (15) using 'bi-gram counts'. For each i and j in $\{1, \dots, k\}$, let $x_{j|i}$ denote the number of times the word j follows word i in the text. Also let $x_i = \sum_{j=1}^k x_{j|i}$ denote the number of times word i appears as the "previous word" in the text.

Then the likelihood (15) is the same as:

$$\prod_{i=1}^k \prod_{j=1}^k p_{j|i}^{x_{j|i}} \quad (16)$$

where $p_{j|i}$ denotes the probability that word j appears as the next word given that the previous word is i .

We can also directly write the likelihood in terms of the bigram counts $(x_{j|i}, j = 1, \dots, k)$ and $i = 1, \dots, k$. The model here is:

$$(x_{j|i}, j = 1, \dots, k) \stackrel{\text{ind}}{\sim} \text{Multinomial}(x_i; p_{j|i}, j = 1, \dots, k).$$

so that the likelihood in terms of the bigram counts is:

$$\prod_{i=1}^k \frac{x_i!}{\prod_{j=1}^k x_{j|i}!} \prod_{j=1}^k p_{j|i}^{x_{j|i}}. \quad (17)$$

The unknown parameters in this model are the probability vectors $(p_{j|i}, 1 \leq j \leq k)$ for each $i = 1, \dots, k$.

[MacKay and Peto \[1995\]](#) perform Bayesian inference with the following prior:

$$(p_{j|i}, j = 1, \dots, k) \stackrel{\text{i.i.d}}{\sim} \text{Dirichlet}(a_1, \dots, a_k)$$

for some $a_1, \dots, a_k > 0$. This means that all the k probability vectors $(p_{j|i}, 1 \leq j \leq k)$ are assumed to be independently distributed according to the same Dirichlet distribution $\text{Dirichlet}(a_1, \dots, a_k)$. The prior density is therefore

$$\prod_{i=1}^k \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \prod_{j=1}^k p_{j|i}^{a_j-1}.$$

The posterior is then:

$$\prod_{i=1}^k \frac{x_i!}{\prod_{j=1}^k x_{j|i}!} \prod_{j=1}^k p_{j|i}^{x_{j|i}} \times \prod_{i=1}^k \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \prod_{j=1}^k p_{j|i}^{a_j-1} \propto \prod_{i=1}^k \prod_{j=1}^k p_{j|i}^{x_{j|i} + a_j - 1}$$

which means that:

$$(p_{j|i}, j = 1, \dots, k) \stackrel{\text{ind}}{\sim} \text{Dirichlet}(x_{j|i} + a_j, j = 1, \dots, k).$$

So the posterior mean estimate of $p_{j|i}$ is:

$$\hat{p}_{j|i} = \frac{x_{j|i} + a_j}{x_i + a_1 + \dots + a_k}.$$

The choice of $a_1, \dots, a_k$ controls the properties of the above posterior. Instead of plugging in some values for $a_1, \dots, a_k$, [MacKay and Peto \[1995\]](#) propose estimating them by maximizing the marginal likelihood of the data ($x_{j|i}, 1 \leq i, j \leq k$) over all values of $a_1, \dots, a_k$.

The marginal distribution of the data given the Dirichlet hyperparameters $a_1, \dots, a_k$ is:

$$\begin{aligned} & P\{x_{j|i}, 1 \leq i, j \leq k \mid a_1, \dots, a_k\} \\ &= \int \prod_{i=1}^k \frac{x_i!}{\prod_{j=1}^k x_{j|i}!} \prod_{j=1}^k p_{j|i}^{x_{j|i}} \times \prod_{i=1}^k \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \prod_{j=1}^k p_{j|i}^{a_j-1} d(p_{j|i} \text{ all } j, i) \\ &= \prod_{i=1}^k \frac{x_i!}{\prod_{j=1}^k x_{j|i}!} \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \int \prod_{j=1}^k p_{j|i}^{x_{j|i}+a_j-1} d(p_{j|i}, 1 \leq j \leq k) \\ &= \prod_{i=1}^k \frac{x_i!}{\prod_{j=1}^k x_{j|i}!} \left[\frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(x_i + a_1 + \dots + a_k)} \prod_{j=1}^k \frac{\Gamma(x_{j|i} + a_j)}{\Gamma(a_j)} \right]. \end{aligned}$$

Remark 5. The normalizing constant of the Dirichlet is explicit in terms of Gamma integrals. This can again be proved by considering connections to the gamma distribution.

In the machine learning literature, the marginal distribution of the data given the hyperparameters of the prior is referred to as the Evidence (see e.g., https://en.wikipedia.org/wiki/Marginal_likelihood). So:

$$\text{Evidence}(a_1, \dots, a_k) = \prod_{i=1}^k \frac{x_i!}{\prod_{j=1}^k x_{j|i}!} \left[\frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(x_i + a_1 + \dots + a_k)} \prod_{j=1}^k \frac{\Gamma(x_{j|i} + a_j)}{\Gamma(a_j)} \right].$$

The Evidence can be maximized over $a_1, \dots, a_k$ to get a good choice of the hyperparameters. Note that the factorial terms do not involve a_j so they can be dropped in this maximization:

$$\text{Evidence}(a_1, \dots, a_k) \propto \prod_{i=1}^k \left[\frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(x_i + a_1 + \dots + a_k)} \prod_{j=1}^k \frac{\Gamma(x_{j|i} + a_j)}{\Gamma(a_j)} \right].$$

It is always better to optimize the logarithm of the evidence (rather than the evidence directly):

$$\begin{aligned} & \log \text{Evidence}(a_1, \dots, a_k) \\ &= \text{constant} + \sum_{i=1}^k \left\{ \log \Gamma(A) - \log \Gamma(x_i + A) + \sum_{j=1}^k [\log \Gamma(x_{j|i} + a_j) - \log \Gamma(a_j)] \right\}. \end{aligned}$$

where $A := a_1 + \dots + a_k$ and $x_i = \sum_{j=1}^k x_{j|i}$. One can compute the gradient with respect to a using the digamma function (see https://en.wikipedia.org/wiki/Digamma_function):

$$\Psi(z) := \frac{d}{dz} \log \Gamma(z).$$

This gives:

$$\frac{\partial}{\partial a_j} \log \text{Evidence}(a_1, \dots, a_k) = \sum_{i=1}^k \{ \Psi(A) - \Psi(x_i + A) + \Psi(x_{j|i} + a_j) - \Psi(a_j) \}.$$

With these gradients (scipy has an inbuilt function for digamma calculation), some numerical optimization routine can be used for maximizing the log-Evidence over $a_1, \dots, a_k$ (see code).

References

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